Advance Python Assignment

Topic-4: Reading and Writing Files

Q1. Explain reading data from a file in Python.

Ans.
In Python, reading data from a file is an important part of file handling. It allows us to access stored information from a file. Python provides functions like read(), readline(), and readlines() to read file data. The read() function reads the entire file, readline() reads one line at a time, and readlines() reads all lines into a list. These functions help in processing file data easily. Reading files is useful in applications where data is stored and needs to be accessed later.

Example:
```
file = open("sample.txt", "r")
data = file.read()
print(data)
file.close()
```

Q2. Explain writing data into a file in Python.

Ans.
Writing data into a file in Python is done using the write() and writelines() functions. These functions help in storing information in a file. The write() function is used to write a single string, while writelines() is used to write multiple lines. If the file is opened in 'w' mode, it will overwrite the existing content, and in 'a' mode, it will add new data at the end. Writing files is useful for saving program results and user data for future use.

Example:
```
file = open("output.txt", "w")
file.write("This is a file writing example")
file.close()
```